

Q1 A college stores the names of students in an ArrayList. The administrator wants to display all student names using an Iterator instead of a for loop. 10 marks

A college stores the names of students in an ArrayList. The administrator wants to display all student names using an Iterator instead of a for loop.

Write a Java program to:

Create an ArrayList of student names.

Read 8 student names from the user.

Traverse the list using an Iterator.

Display each student name on a separate line.

Display the total number of students stored in the list.

Your Code

```
1 import java.util.*;
2 public class main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         ArrayList<String>students=new ArrayList<String>();
6         for(int i=0;i<8;i++){
7             if(sc.hasNextLine()){
8                 String name=sc.nextLine().trim();
9                 students.add(name);
10            }
11        }
12        Iterator<String>it=students.iterator();
13        while(it.hasNext()){
14            System.out.println(it.next());
15        }
16        System.out.println("total students: "+students.size());
17        sc.close();
18    }
19 }
```

Run

Save

Input

```
sri
kanth
ink
thuk
```

Output

```
ink
thuk
pak
nigh
total students: 6
```

Q2 Remove Even Numbers Using Iterator

10 marks

A company maintains a list of employee ID numbers. The management wants to remove all employees whose ID numbers are even, using an Iterator.

Write a Java program to:

Create an ArrayList<Integer>.

Read 10 employee ID numbers from the user.

Traverse the list using an Iterator.

Remove all even ID numbers using the remove() method of the Iterator.

Display the updated list containing only odd employee IDs.

```
ne_java_compiler

Main.java
1 import java.util.*;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         ArrayList<Integer> employee = new ArrayList<Integer>();
6         for (int i = 0; i < 10; i++) {
7             int id = sc.nextInt();
8             employee.add(id);
9         }
10        Iterator<Integer> it = employee.iterator();
11        while (it.hasNext()) {
12            int id = it.next();
13            if (id % 2 == 0) {
14                it.remove();
15            }
16        }
17        for (Integer id : employee) {
18            System.out.print(id + " ");
19        }
20    }
21 }
```

input

```
123654
78961
78542
45879
12345 12547 78961 45879

...Program finished with exit code 0
Press ENTER to exit console.
```

Q3 Product Price Analysis Using Iterator

10 marks

A supermarket stores the prices of various products in an `ArrayList<Double>`. The manager wants to analyze the prices using an Iterator.

Write a Java program to:

- Create an `ArrayList<Double>`.
- Read the prices of 10 products from the user.
- Traverse the list using an Iterator.
- Calculate and display:
 - Total price of all products.
 - Average product price.
 - Highest product price.
- Display all product prices using the Iterator.

Your Code

```
1 import java.util.*;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         ArrayList<Double> prices = new ArrayList<>();
6         System.out.println("Enter prices of 10 products:");
7         for (int i = 0; i < 10; i++) {
8             System.out.println("Price " + (i + 1) + ": ");
9             prices.add(sc.nextDouble());
10        }
11        Iterator<Double> it = prices.iterator();
12        double total = 0;
13        double highest = Double.NEGATIVE_INFINITY;
14        System.out.println("\nProduct prices (using Iterator):");
15        while (it.hasNext()) {
16            double p = it.next();
17            System.out.print(p + " ");
18            total += p;
19            if (p > highest) highest = p;
20        }
21        double average = total / prices.size();
22        System.out.println("\n\nTotal price: " + total);
23        System.out.println("Average product price: " + average);
24        System.out.println("Highest product price: " + highest);
25        sc.close();
26    }
27 }
```

Input

8
9
6
3

Output

Product prices (using Iterator):
5.0 6.0 3.0 8.0 9.0 7.0 8.0 9.0
6.0 3.0

Total price: 64.0
Average product price: 6.4
Highest product price: 9.0